AI Agent Integration Reflection

To improve my Automated Pet Feeder project, I used Microsoft Copilot to refine the feeding process and explore hardware integration. I asked for advice on error handling and alerts. Copilot suggested multiple monitoring intervals, dynamic thresholds for small pets, and alerts for servo malfunctions. These ideas revealed weaknesses in my original design, especially in detecting partial feedings and mechanical problems.

It also proposed alternative solutions, such as adaptive feeding thresholds, repeated weight checks, servo feedback monitoring, camera or motion sensors, cloud-based logging, mobile notifications, battery backup, and using machine learning to optimize feeding schedules. These suggestions helped me understand how to make the system more reliable and suitable for shelters with pets of different sizes and activity levels.

For building it with actual hardware, Copilot suggested that the system could use platforms like Arduino or Raspberry Pi. These devices support servo motors, weight sensors, and real-time clocks, allowing scheduled feeding and error detection. With basic coding and sensor integration, the prototype can develop into a reliable, low-cost automated pet feeder.

I also reflected on ethics and limitations. Although Copilot gave useful suggestions, AI cannot replace human judgment, and automation errors could harm pets if not monitored. Ethical use requires careful testing, supervision, and clear understanding of AI limits.

Overall, Copilot acted like a collaborative reviewer, pointing out design flaws, offering improvements, and encouraging critical thinking. This experience strengthened my solution, improved my understanding of automated systems, and highlighted the importance of combining AI support with ethical responsibility.

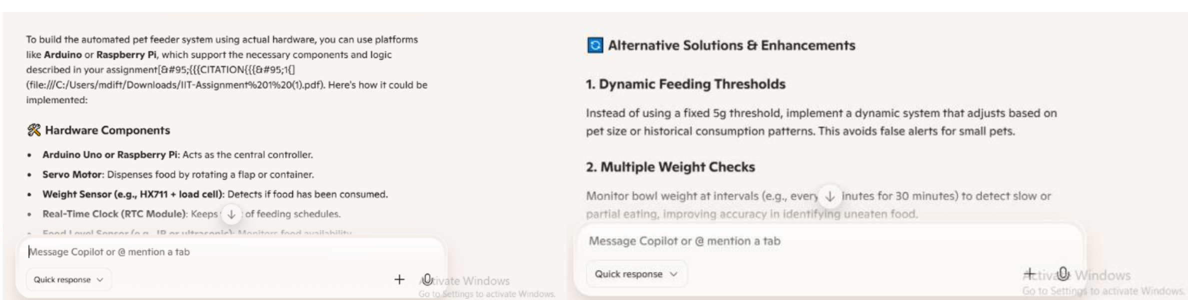

Figure 3: Screenshots